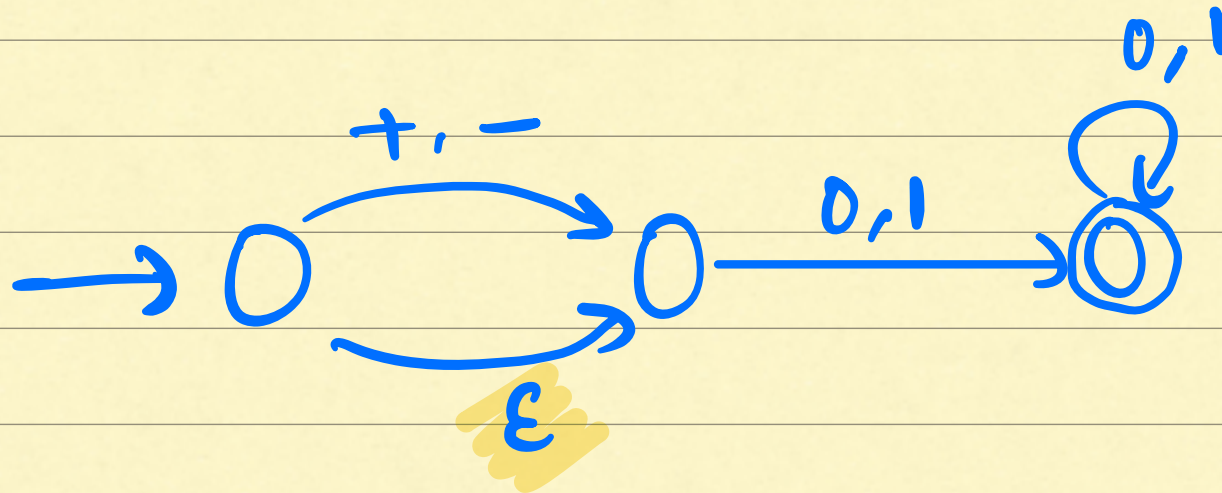


ξ - transitions

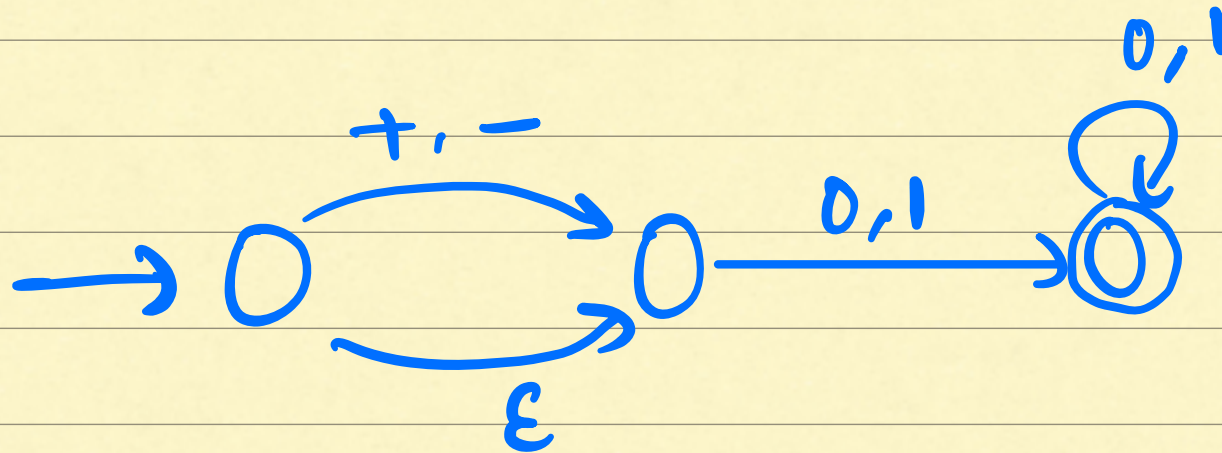
06/08/2026



+ 01001

- 101

+ - 1100



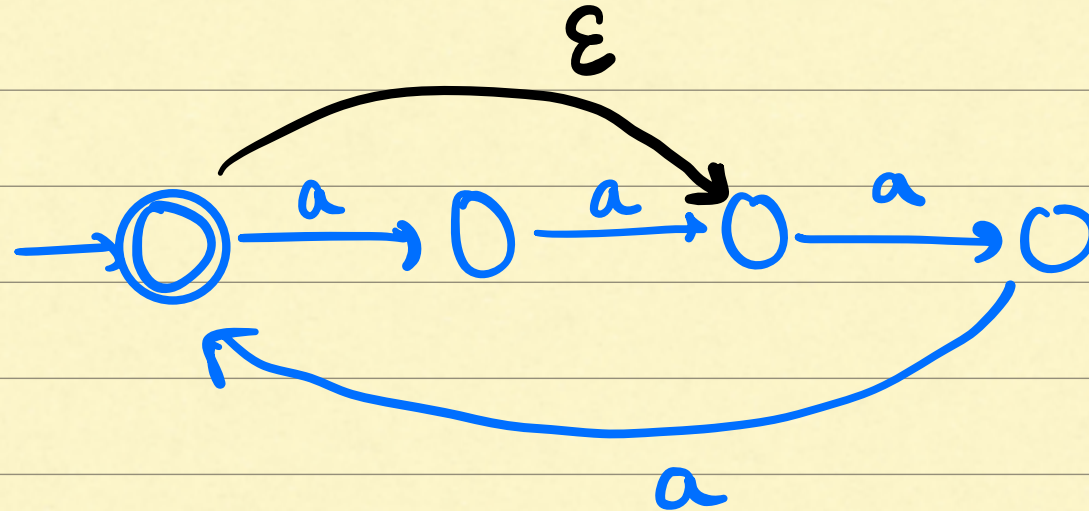
+ 01001

- 101

+ - 1100

01001

another example



What is the language of this automaton?

... one more feature ...

We now allow an automaton to

jump

from one state to another

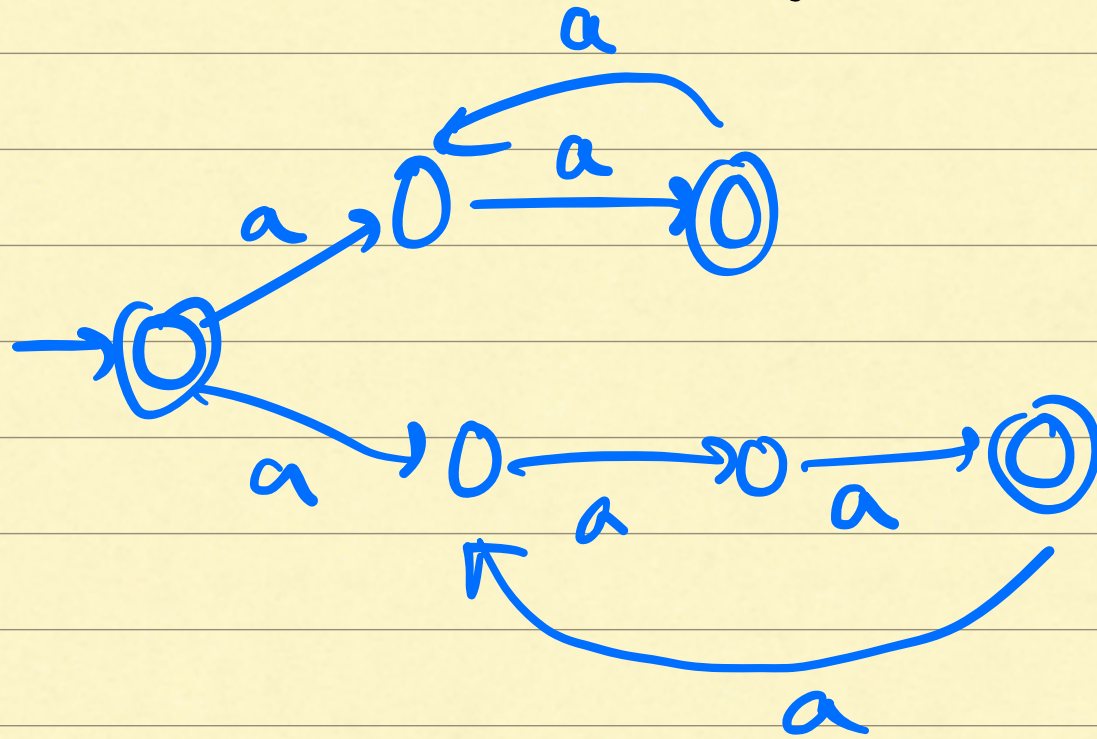
Defⁿ (ϵ -NFA): An ϵ -NFA is a tuple

$$(Q, \Sigma_{\epsilon}, \delta_{\epsilon}, q_0, F)$$

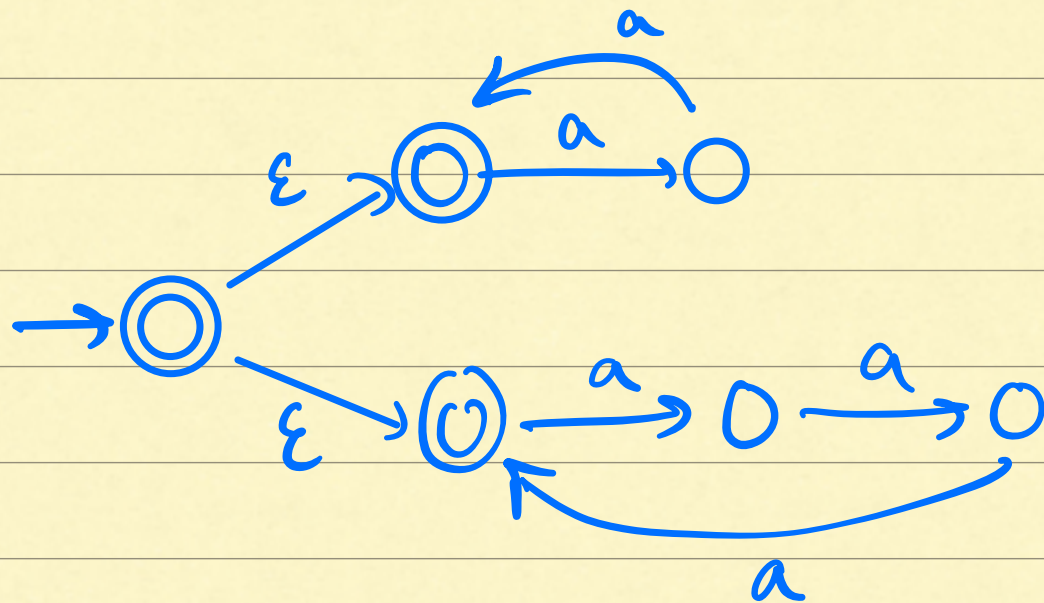
$$\Sigma_{\epsilon} = \Sigma \cup \{\epsilon\}$$

$$\delta_{\epsilon} : Q \times \Sigma_{\epsilon} \rightarrow 2^Q$$

Q: Draw an DFA/NFA That accepts the language $\{w \in \{a\}^* \mid |w| \text{ is either divisible by 2 or by 3}\}$



Q: Draw an ϵ -NFA That accepts the language $\{w \in \{a\}^* \mid |w| \text{ is either divisible by 2 or by 3}\}$



Can you generalize this construction to recognize $L_1 \cup L_2$?

Given two regular languages L_1, L_2 ,

can you draw an ϵ -NFA for $L_1 \circ L_2$?

$$L_1 \circ L_2 = \{ w_1 w_2 \mid w_1 \in L_1 \text{ and } w_2 \in L_2 \}$$

Given two regular languages L_1, L_2 ,

can you draw an ϵ -NFA for $L_1 \cdot L_2$?

$$L_1 \cdot L_2 = \{ w_1 w_2 \mid w_1 \in L_1 \text{ and } w_2 \in L_2 \}$$

Can we always draw a DFA for $L_1 \cdot L_2$?

Does allowing ϵ -transitions
in NFA increase the power of NFA/DFA?

Does allowing ϵ -transitions
in NFA increase the power of NFA/DFA?

! NO !

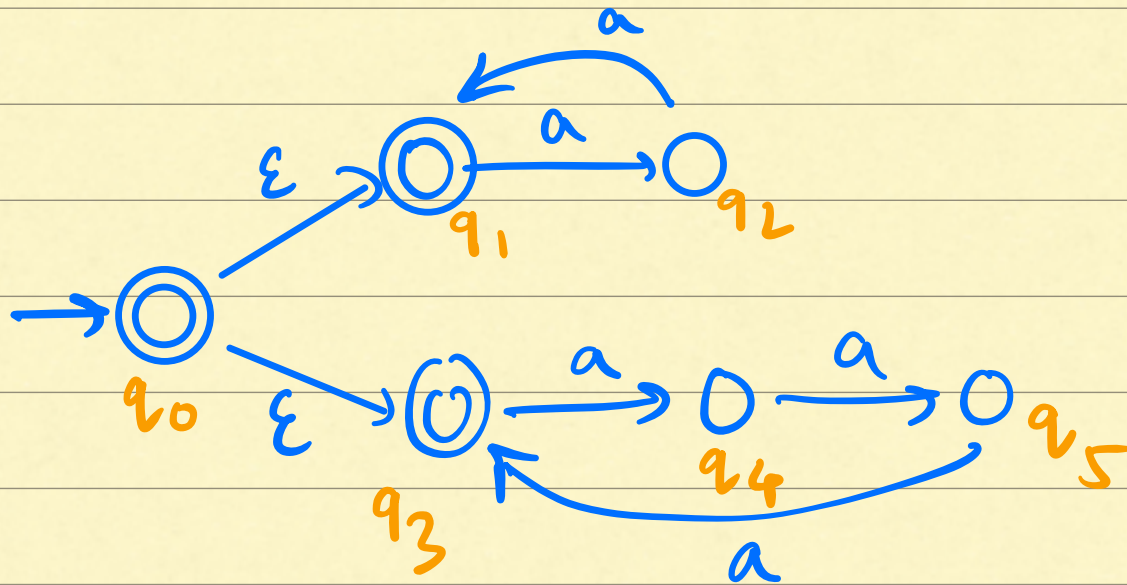
... ϵ -closure ...

Given a set of states $S \subseteq Q$ of an ϵ -NFA,

the ϵ -closure of S , denoted $E(S)$, is

defined as: $\{ q \in Q \mid \exists s \in S. s \xrightarrow{\epsilon} \dots \xrightarrow{\epsilon} q \}$
only ϵ transitions

ϵ -closure



$$E(\{q_0\})$$

$$= \{q_0, q_1, q_3\}$$

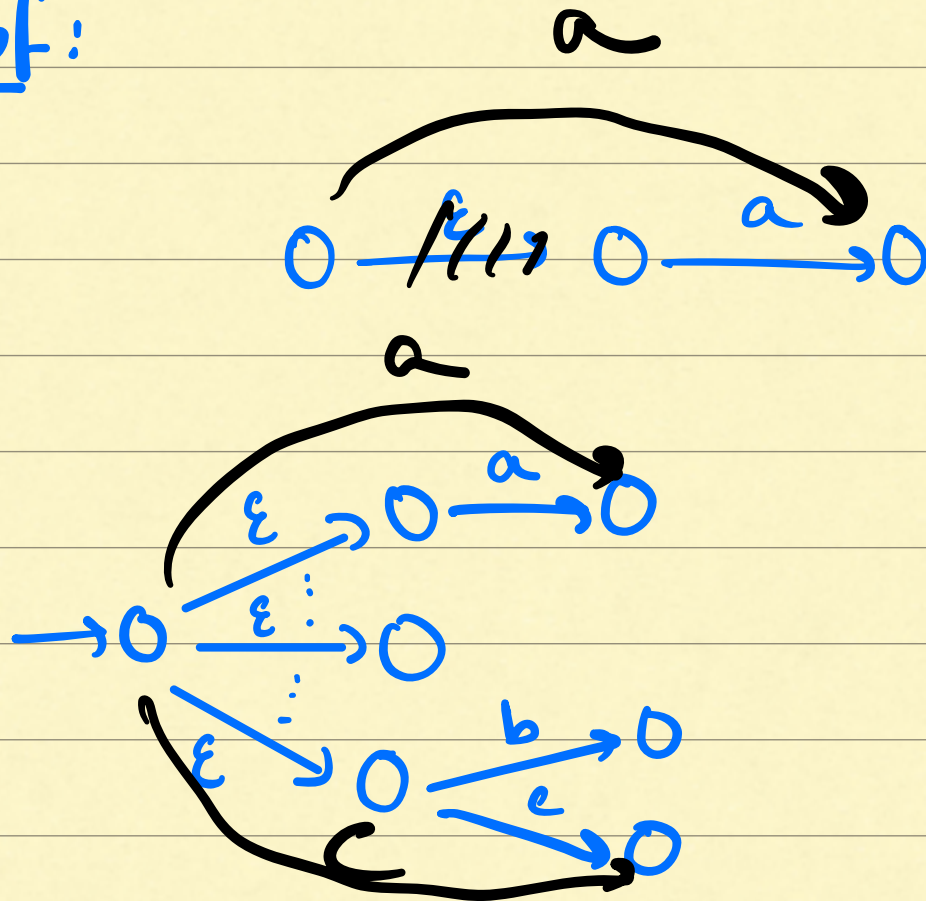
$$E(\{q_3, q_4\})$$

$$= ?$$

... Removing ϵ -transitions ...

Thm: For every ϵ -NFA A_ϵ , there exists an NFA A s.t. $L(A) = L(A_\epsilon)$.

Proof:



Let $A_\epsilon = (Q, \Sigma_\epsilon, \delta_\epsilon, q_0, F)$ be an ϵ -NFA.

consider $A = (Q, \Sigma, \delta, q_0, F)$ be an NFA

defined as $\delta(S, a)$

$$= \{ q \in Q \mid \exists s \in S. q \in E(s) \}$$

claim: $L(A_\epsilon) = L(A)$

To prove $L(A_\epsilon) = L(A)$

Proof: Let $w \in L(A_\epsilon)$

$\Rightarrow \exists$ a run of A_ϵ on w starting at q_0 ,
and terminating at $q_f \in F$.

$$q_0 \rightsquigarrow q_1 \xrightarrow{\epsilon} q_2 \rightsquigarrow q_3 \xrightarrow{\epsilon} q_4 \rightsquigarrow q_f$$

Then we can prove that an "equivalent"
run exists in A , using induction on
the no. of ϵ -transitions present in the
above run.